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Biology
Class 10

Sindh Textbook Board

(English Medium)

Based on the Sindh Textbook Board Book

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Gaseous Exchange

Unit 2
Homeostasis

Unit 3
Coordination

Unit 4
Support and Movement

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Reproduction

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Man and His Environment

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Gaseous Exchange

UNIT 1

Gaseous Exchange in Plants

LESSON 1

1. What is gaseous exchange?

Gaseous exchange means the exchange of gases like oxygen and carbon dioxide between an organism and the environment. This exchange takes place in two life processes: in photosynthesis as well as in respiration.

2. How do plants exchange gases?

Plants carry out gaseous exchange through the root, the stomata and the stem; all these parts take part in gaseous exchange.

- Leaves: mainly take in carbon dioxide and release oxygen through the stomata.
- Stem: exchanges gases through lenticels.
- Root: the root also exchanges gases.

3. Stomata and guard cells

A stoma is a small pore in a leaf. Each stoma is surrounded by two guard cells. The guard cells help in opening and closing the stoma.

4. Why are stomata open during the daytime?

During the day carbon dioxide is needed for photosynthesis, so the stomata stay open so that carbon dioxide can enter and oxygen can leave.

5. Other organisms and the body surface

Some organisms exchange gases through the whole body surface without special organs, e.g. the earthworm, the frog (through its skin), and single-celled organisms like Amoeba.

6. Experiment: carbon dioxide in respiration

Germinating seeds are placed in a flask and the air passing from it is passed through lime water → the lime water turns milky → this proves that carbon dioxide is released during respiration.

Gaseous Exchange in Humans: Pathway and Lungs

LESSON 2

1. The human respiratory pathway

In humans the pathway of air is in this order:

Nose → pharynx → larynx → trachea → bronchi → bronchioles → alveoli

Air enters through the nose, passes through the pharynx and larynx into the trachea, then through the bronchi and bronchioles to the alveoli, where gaseous exchange takes place.

2. Why is breathing through the nose better?

The nose has hairs and mucus which trap dust and germs and clean the air, and warm and moisten it; the mouth does not provide this protection, so breathing through the nose is better than breathing through the mouth.

3. Where is the larynx?

The larynx is present at the start of the trachea. It comes after the pharynx and before the trachea in the pathway.

4. Respiratory surface and alveoli

The respiratory surface in humans is the alveoli, i.e. the air sacs. Real gaseous exchange takes place at the surface of the alveoli, not in the nose, trachea or bronchioles — those are only the pathway.

5. Features of a good respiratory surface

- It should be thin and moist.
- It should be permeable, i.e. gases can pass through it easily.
- It should be very large, so that more gas can be exchanged.

When all these features are present together, the exchange is fast and efficient.

The Process of Breathing

LESSON 3

1. Difference between three terms

Term	Meaning
Breathing (ventilation)	Movement of air in and out
Gaseous exchange	Swapping of gases between the body and the environment
Respiration	Chemical process in cells in which food is broken down to release energy

2. Breathing in — inspiration

- The intercostal muscles contract.
- The diaphragm contracts.
- The ribs move outward and upward, not inward.
- The chest capacity increases and air rushes in.

Order: muscles contract → chest capacity increases → air rushes in

3. Breathing out — expiration

In exhalation (expiration) the muscles relax, the chest capacity decreases and air is pushed out.

4. Inhaled and exhaled air

Air	Oxygen	Carbon dioxide
Inhaled	More	Less
Exhaled	Less	More, and it is warm and moist

5. Why does the breathing rate rise?

A rise in carbon dioxide in the blood increases the breathing rate. During or after exercise breathing also becomes faster, because the muscles need more energy, which needs more oxygen and produces more carbon dioxide to be removed.

6. Oxygen debt

During hard exercise the muscles work without enough oxygen and lactic acid is formed; the extra oxygen needed afterwards to remove that lactic acid is called oxygen debt.

Breathing Disorders and Effects of Smoking

LESSON 4

1. Emphysema

This disorder is linked to damage of the alveoli. In emphysema the walls of the alveoli break down and breathing becomes difficult.

2. Asthma

This disorder is linked to narrowing of the air passage. In asthma the air passage narrows and breathing becomes difficult.

To protect an asthma patient: avoid dust, smoke, cold air and other triggers, take medicine on time, live in a clean environment and follow the doctor's instructions.

3. Lung cancer

Lung cancer is the uncontrolled growth of cells in the lungs. It is mostly caused by smoking and makes breathing difficult.

4. Why is smoking harmful?

Cigarette smoke contains poisonous substances like nicotine, tar and carbon monoxide which damage the lungs and the air passage.

- Disorders connected with smoking: bronchitis, emphysema, asthma and lung cancer.

5. Prevention of respiratory disorders

- Avoid smoking.
- Live in clean air and stay away from dust and smoke.
- Exercise regularly and eat a balanced diet.
- Get vaccinated.
- Seek timely treatment if illness occurs.

Homeostasis

UNIT 2

Introduction to Homeostasis and Homeostasis in Plants

LESSON 5

1. What is homeostasis?

Keeping the internal condition of an organism steady is called homeostasis. In other words, the combined biological processes that keep the internal environment stable are homeostasis.

2. Why does the body need homeostasis?

Homeostasis is needed to keep the body's internal environment (temperature, water, salts and pH) steady so that the cells and organs can work properly.

3. Removal of extra water in plants

- Transpiration: extra water leaves as vapour through the leaves.
- Guttation: water leaves as drops from the leaf edges.
- Excretion: plants also remove extra water by excretion.

Remember: the removal of extra water from the body in the form of vapour is called transpiration, not guttation.

4. Halophytes — plants of salty places

Plants growing in salty places like the seashore are halophytes, i.e. salt-tolerant plants.

They survive in salty water in these ways:

- By storing salt in their cells.
- By removing extra salt.
- By keeping their water balance steady.

Homeostasis in Animals and Humans

LESSON 6

1. Skin — the largest organ

The skin is considered the largest organ of the human body. The skin is not just a covering; it also plays a big part in homeostasis.

2. What is thermoregulation?

Keeping the body temperature steady at a set level is called thermoregulation, and the process that keeps the temperature steady is also thermoregulation.

3. Layers of the skin and the dermis

The layer of the skin containing a network of blood vessels is called the dermis. The dermis contains the network of blood vessels, nerves and sweat glands.

4. Why is the skin an excretory organ?

The skin removes extra water, salts and waste substances like urea from the body through sweat, so it is considered an excretory organ.

5. What does the skin do in heat?

- The blood vessels of the skin widen.
- More sweat is produced.
- The evaporation of sweat cools the body and the temperature stays steady.

6. What does the skin do in cold?

In cold the skin narrows the blood vessels and reduces sweating to conserve heat. In this way the skin works as a thermoregulatory organ: it cools the body in heat and conserves heat in cold.

Human Kidney and Structure of the Nephron

LESSON 7

1. The kidney and its membrane

The kidney is enclosed in a membrane called the peritoneum.

2. Parts of the kidney and their work

Part	Work / place
Cortex	Outer part
Medulla	Inner part
Pelvis	Urine-collecting part
Nephron	Unit that filters the blood and makes urine
Ureter	Tube carrying urine to the bladder

3. The human urinary system

The human urinary system consists of two kidneys, two ureters, one bladder and one urethra.

Pathway: the kidneys make urine → the ureters carry it to the bladder → the bladder stores it → it leaves through the urethra.

4. Nephron — the unit of the kidney

The nephron is the unit of the kidney. It includes the glomerulus, Bowman's capsule, the proximal convoluted tubule, the loop of Henle, the distal convoluted tubule and the collecting duct. Its detailed functions are given in the next lesson.

Functions of the Nephron and Formation of Urine

LESSON 8

1. Structure of the nephron again

The nephron consists of the glomerulus (a ball of blood capillaries), Bowman's capsule, the proximal convoluted tubule, the loop of Henle and the distal convoluted tubule; the collecting duct is joined with it.

2. Steps of urine formation

Filtration of blood in the glomerulus → selective reabsorption in the tubule → tubular secretion → collection of urine in the collecting duct

3. Selective reabsorption

In the nephron, selective reabsorption takes place in the convoluted tubule. By reabsorption, useful substances like water, glucose and salts return to the blood.

4. Tubular secretion

By tubular secretion, extra and harmful substances move from the blood into the tubule so that they leave in the urine.

Process	What happens?
Reabsorption	Useful substances: water, glucose, salts return to the blood
Secretion	Extra and harmful substances from the blood into the tubule, then out with the urine

5. Role of the blood-vessel network

The glomerulus (a ball of blood capillaries) filters the blood; the network of blood vessels around the tubule takes reabsorbed substances back into the blood and supplies substances for secretion; thus the blood network plays the central role in making urine.

6. ADH hormone

The ADH hormone is secreted by the pituitary gland.

Disorders of the Excretory System and Kidney Stones

LESSON 9

1. Kidney diseases

- Kidney stones
- Nephritis — inflammation of the kidney
- Urinary tract infection
- Kidney failure

2. Methods of treatment

- Medicines
- Breaking up stones — lithotripsy — or surgery
- Dialysis
- Kidney transplant

3. Why is dialysis needed?

When the kidneys fail and cannot remove waste substances and extra water from the blood, dialysis is needed to clean the blood with a machine.

Remember: dialysis does the kidney's work by a machine, in a transplant the kidney is replaced, and lithotripsy is for breaking stones — the three treatments are for different conditions.

Coordination

UNIT 3

Types of Coordination and the Neuron

LESSON 10

1. What is coordination?

Coordination means the parts of the body working together in an orderly way. A change in the environment is called a stimulus, and the movement made in answer to that change is called a response.

- Stimulus: a change in the environment.
- Receptor: the part that perceives the stimulus.
- Response: the movement made in answer to the stimulus.
- Effector: the part that responds, such as a muscle.

2. Two types of coordination

Type	Method	Speed and duration
Nervous	Through electrochemical signals	Fast, for a short time
Chemical	Through hormones, by the blood	Slow, long-lasting

A hormone is a chemical released from one cell that signals another cell at a distance, or is sent through the blood.

3. Why is nervous coordination faster?

Nervous coordination travels directly and fast through electrochemical signals, while in chemical coordination hormones travel slowly through the blood.

4. Coordination in plants

Coordination in plants takes place chemically, through hormones. Plants have no nervous system.

5. Which system is better?

Both systems are necessary in their place; the nervous system acts fast and briefly, while the chemical (endocrine) system acts slowly but for a long time, so together they work best.

6. Feedback system

A feedback system is one in which the result of an action is reported back to the system so that the action can be increased or decreased and balance is maintained.

Note: three types of neurons are named in this chapter's questions — the sensory neuron, the intermediate neuron and the motor neuron — and they are explained in detail in the reflex action lesson. The signal passes from the receptor through the sensory neuron to the spinal cord and then through the motor neuron to the effector.

Human Nervous System and Its Parts

LESSON 11

1. Main parts of the human brain

Part	Work
Cerebrum	Largest part, thinking and memory
Cerebellum	Balance and precision of movement
Brain stem (medulla, pons, midbrain)	Basic actions like breathing and heartbeat

The brain is covered by three meninges membranes and has ventricles inside.

2. Frontal lobe

The frontal lobe of the cerebrum is connected with reasoning, thinking and decision-making.

3. Parts of the nervous system

- Central Nervous System (CNS)
- Peripheral Nervous System (PNS)
- Somatic system
- Autonomic system — controls survival actions like heartbeat and breathing.

4. Why do people differ in intelligence?

Intelligence depends not only on the parts of the brain but on the number of neurons, the connections between them, the growth of the brain, education and environment; differences in these cause differences in intelligence.

Reflex Action and Reflex Arc

LESSON 12

1. What is a reflex action?

A reflex action is the quick response that takes place without the brain thinking. When a stimulus comes, the body moves at once.

2. Reflex arc — the pathway of the signal

Stimulus → receptor → sensory neuron → spinal cord → motor neuron → effector (muscle) → immediate movement

- The stimulus is felt by the receptor.
- The signal travels through a sensory neuron to the spinal cord.
- From there it travels through a motor neuron to the effector (muscle).
- The muscle moves at once, without the brain having to think.

3. How many neurons?

A reflex action usually involves three neurons: sensory neuron, intermediate neuron and motor neuron. The answer given in the book is at most three neurons.

Human Receptors: Eyes and Ears

LESSON 13

1. Parts of the human eye

Part	Description
Sclera	White outer coat
Cornea	Transparent front part
Iris	Coloured part
Pupil	Opening
Lens	Convex lens
Vitreous	Part behind the lens
Retina	Part with rod and cone cells

2. How is the image formed?

Light passes through the cornea and lens and forms an image on the retina → the signal travels to the brain through the optic nerve.

3. Lens and accommodation

The human eye has a convex lens which focuses light on the retina. The adjustment of the eye lens for near and far objects is called accommodation.

4. Photoreceptor cells

Photoreceptor cells are present in the retina. The retina of the eye contains photoreceptor cells called rods and cones.

5. Vitamin A and vision

Vitamin A is essential for eye health and correct vision.

6. Colour blindness

A colour-blind person usually cannot distinguish green from red. The reason is that in such a person's retina some cone cells are defective or absent, so he cannot distinguish those colours.

7. A note about the ear

In the questions of this chapter, the name cochlea appears as an option in connection with the ear, where it is made clear that photoreceptor cells are in the retina, not in the cochlea. Detailed structure of the ear is not given in this chapter's questions.

Endocrine System and Important Glands

LESSON 14

1. What is the endocrine system?

The endocrine system is the system of glands that release hormones into the blood. Hormones reach the target organs through the blood and act there.

2. How does blood glucose stay steady?

- When blood glucose rises, the pancreas releases insulin, which lowers it.
- When glucose falls, it releases glucagon, which raises it.

- In this way the level stays steady — this is an example of feedback.

3. Islets of Langerhans

The islets of Langerhans in the pancreas make the hormones insulin and glucagon, which control the blood glucose level.

4. Thyroid and goitre

Due to iodine deficiency the thyroid gland swells, which is called goitre.

5. Emergency hormones

The emergency hormones are adrenaline and noradrenaline, called the hormones of the adrenal glands. They prepare the body in times of danger.

6. FSH and LH

Gonads (sex glands) are the target organs of both FSH and LH — both hormones act on the gonads.

7. Diabetes (sugar)

Diabetes (sugar) is the disease in which blood glucose rises due to lack of insulin or its failure to act.
Prevention: a balanced diet, exercise, keeping weight under control and regular check-ups.

Nervous Disorders

LESSON 15

1. Paralysis

Possible causes of paralysis:

- Bursting or blockage of a blood vessel in the brain.
- Brain injury.
- High blood pressure.
- Nerve damage.

2. Epilepsy

Epilepsy is a disorder of the brain in which sudden abnormal electrical activity in the brain causes fits of jerking movements and unconsciousness.

3. Difference to remember

Disorder	Main cause / sign
Paralysis	Bursting or blockage of a blood vessel, injury, high blood pressure, nerve damage
Epilepsy	Sudden abnormal electrical activity, fits of jerking and unconsciousness

Both disorders are connected with the brain and nerves, but their causes and signs are different. Besides these, the other disorders that appear in this chapter are colour blindness, connected with defective cone cells, and goitre, connected with the thyroid gland.

Support and Movement

UNIT 4

Concept of Support and Movement and Types of Bones

LESSON 16

1. What is movement?

When an organism feels restlessness due to a stimulus, i.e. an outside effect, that restlessness is removed by movement. Organisms move to remove the restlessness.

2. What is a skeleton?

The framework that gives a structure a firm shape is called the skeleton. The skeleton is the framework of bones and it gives the body a firm shape. The human skeleton is made of 206 bones.

3. Important functions of the skeleton

- Giving the body shape and support
- Protecting the internal organs, such as the brain, heart and lungs
- Helping movement together with the muscles
- Making blood cells
- Storing minerals like calcium

The strength of the skeleton is due to the hard structure of bones, so the skeleton is called a strong system.

4. Difference between bone and cartilage

Bone	Cartilage
Hard and strong	Soft and flexible
Has blood vessels	Has no blood vessels

5. Cells and places of cartilage

The cell of cartilage is called a chondrocyte. These cells are quite different from bone cells. Cartilage is found in the places where flexibility is needed.

- Cartilage is present in the ears
- Cartilage is present in the tip of the nose
- Cartilage is present in the rings of the trachea
- Cartilage is present on the surfaces of joints
- Cartilage is present at the ends of the ribs

The Human Skeleton: Axial and Appendicular Bones

LESSON 17

1. The human skeleton

The human skeleton is a strong framework of 206 bones. It gives the body shape and support and produces movement together with the muscles. Its two big parts must be understood: the axial skeleton and the appendicular skeleton.

2. Sacrum and pelvic girdle

The sacral vertebrae join with the pelvic girdle. This joint helps to connect the legs with the central part of the body.

3. Bones of the leg and foot

Each leg and foot has these bones:

- Femur: the thigh bone, known as the largest bone of the leg
- Tibia and fibula: the two shin bones
- Patella: the knee cap
- Tarsals: 7 ankle bones
- Metatarsals: 5 bones
- Phalanges: 14 toe bones

Bone or group	Place and number
Femur	In the thigh, one
Tibia and fibula	In the shin, two
Patella	In the knee, one
Tarsals	In the ankle, 7
Metatarsals	In the foot, 5
Phalanges	In the toes, 14

4. A way to remember

Remember the leg from top to bottom: femur → patella → tibia and fibula → tarsals → metatarsals → phalanges. The numbers and names are often asked in exams.

Joints and Muscle Movement

LESSON 18

1. What is a joint and its types

The place where two bones meet is called a joint. There are three big types in the human body:

- Fixed or immovable joints: do not move at all, e.g. between the bones of the skull and between the teeth and the jaws
- Cartilaginous or slightly movable joints: slight movement takes place, e.g. between the ribs and the vertebrae
- Synovial or freely movable joints: free movement takes place

2. Types of synovial joints

Joint	Movement	Example
Hinge joint	Movement in one direction	Knee and elbow
Ball-and-socket joint	Movement in all directions	Shoulder and hip
Pivot joint	Rotating movement	Neck
Gliding joint	Sliding movement	Wrist and ankle

In a ball-and-socket joint the round head (ball) of one bone fits into the cup-like hollow (socket) of another bone, so the shoulder and hip joints can move in every direction.

3. Ligament and tendon

Ligament	Tendon
Joins bone to bone	Joins muscle to bone
Is a strong band at a joint	Carries the muscle's pull to the bone

4. Muscle movement: flexion and extension

Flexion → a muscle contracts and the joint bends, e.g. bending the arm. Extension → the opposite muscle contracts and the joint straightens, e.g. straightening the arm. The movement that straightens a joint is called extension.

5. Antagonist muscles

The pair of muscles working in opposite directions on one bone is called the antagonist pair. Biceps and triceps are its best example:

- Biceps: the muscle with two heads, bends the arm
- Triceps: the muscle with three heads, straightens the arm
- The big muscle of the thigh is the quadriceps (tetraceps)

When one muscle contracts the other relaxes, so movement is smooth.

Disorders of the Skeletal System

LESSON 19

1. Bone disorder: rickets

Rickets is a disease of children in which the bones become soft and weak. A healthy bone is hard and strong, but in rickets the bone loses its hardness.

2. Cause of rickets

- Deficiency of vitamin D is the biggest cause
- Deficiency of calcium is also a cause
- Deficiency of phosphorus is also a cause

Softening of children's bones due to vitamin D deficiency is called rickets. Calcium and phosphorus are the minerals needed for the strength of bones, and the skeleton also stores calcium.

3. Difference from other names

Name	Note
Rickets	Softening of bones in children due to vitamin D deficiency
Osteoporosis	A different name from rickets, not the children's soft-bone disease
Osteoarthritis	A different name from rickets, not the children's vitamin D deficiency disease

Remember: if a question has children, soft bones and vitamin D deficiency together, the answer is always rickets. To prevent it, the deficiency of vitamin D, calcium and phosphorus must be made up.

Reproduction

UNIT 5

Sexual Reproduction in Plants

LESSON 20

1. Sexual reproduction in plants and the flower

Reproduction is the process necessary for the survival of organisms and the continuation of life. In sexual reproduction a new organism is formed by the fusion of male and female gametes, and new combinations are produced in it, which are necessary for evolution.

2. Parts of an angiosperm flower

- Sepals: together form the calyx
- Petals: together form the corolla
- Stamen: the male part, with anther and filament
- Carpel: the female part, with stigma, style and ovary

3. Pollination

Pollination is the transfer of pollen grains from the anther to the stigma. The series of the process is: anther → pollen grain → stigma → fertilisation → seed and fruit. Characteristics of insect-pollinated plants:

- Flowers are colourful
- Flowers are fragrant
- Flowers are nectar-bearing, i.e. they have nectar
- They have sticky pollen

4. Male and female gametophytes

The male gametophyte is the pollen grain. It has an outer coat (exine), an inner coat (intine), a tube cell and a generative cell; the generative cell forms two male gametes. The female gametophyte is the embryo sac.

5. Structure of the ovule

- Outer coats: integuments
- Micropyle: a small opening
- Nucellus: the inner part
- Embryo sac: the female gametophyte

6. Double fertilisation and endosperm

In angiosperms the endosperm is formed from the 3N zygote, i.e. triple fusion, which feeds the seed. The seed coat and the embryo are separate parts from it; the cotyledon is also separate from the endosperm.

7. Life cycle and seed germination

The life cycle of angiosperms: seed → germination → plant → flower → pollination → fertilisation → seed and fruit → seed again. Seed germination is the process in which a new plant emerges from the seed; it needs water, suitable temperature and oxygen.

Epigeal	Hypogeal
Seed leaves come above the ground	Seed leaves stay in the ground

Asexual Reproduction in Plants

LESSON 21

1. Asexual and vegetative reproduction

In asexual reproduction a new organism is formed from a single parent without gametes. Vegetative propagation is its special form in which a new plant is formed from a part of the plant such as the stem, root or leaf.

Sexual reproduction	Asexual reproduction	Vegetative propagation
A new organism is formed by the fusion of male and female gametes	A new organism is formed from one parent without gametes	A new plant is formed from a part of the plant (stem, root, leaf)

2. Methods of asexual reproduction in plants

- Runner: a stem spreading on the ground
- Rhizome: an underground stem like ginger
- Tuber: a thick stem like potato
- Bulb: a round structure like onion
- Sucker: a stem arising from the root region of the mother plant
- Budding, spores and tissue culture are also methods

Mint reproduces vegetatively by a stem creeping on the ground with little nourishment; this is an example of the runner method. Single-celled organisms also reproduce asexually without fusion by means of spores.

3. Apomixis

The method of seed production without male-female fusion is apomixis. In it a seed is formed but the fusion of fertilisation does not take place. Parthenocarpy and parthenogenesis are different terms from it.

4. How does a new plant obtain food?

At first a new plant gets food stored in the parent plant or the seed, then it takes in water and minerals through the roots and makes its own food by photosynthesis in the leaves.

Reproduction in Animals

LESSON 22

1. Introduction to reproduction in animals

Reproduction in animals is both sexual and asexual. In sexual reproduction male and female gametes fuse, while in asexual reproduction new organisms are formed from a single organism without fusion.

2. Methods of asexual reproduction in animals

Method	Example	Explanation
Binary fission	Amoeba	One organism divides into two parts and forms two new organisms
Budding	Hydra	A small outgrowth on the body becomes a new organism
Fragmentation	Planaria	A new organism is formed from a piece of the body
Parthenogenesis	The fourth method in the source	A method of reproduction without fusion

Remember that plant methods like runner, rhizome and bulb do not apply to animals. Remember four names for animals: binary fission, budding, fragmentation and parthenogenesis, together with their examples.

Note: the source has few facts for this lesson — all four methods are given in one detailed answer, so the whole lesson is built around these four methods and their examples.

Sexual Reproduction in Humans: Male and Female Reproductive Systems

LESSON 23

1. Sexual reproduction in humans

Sexual reproduction in humans takes place by the fusion of male and female gametes. The male gamete is the sperm and the female gamete is the ovum. The organs and glands of both systems must be understood separately.

2. Male reproductive system

- Testes: the male gonads, present in the scrotal sac
- Epididymis: a part joined with the testis
- Vas deferens: the tube carrying sperms
- Seminal vesicle: a gland
- Prostate gland: a gland
- Cowper's glands: glands
- Penis: the external organ

The male gonad, i.e. the testis, is located in the scrotal sac.

3. Spermatogenesis

The process of sperm formation is called spermatogenesis. The series: seminiferous tubules of the testes → primary spermatocyte → meiosis → four haploid spermatids → mature sperms, i.e. from one primary spermatocyte four sperms are formed.

4. Female system and ovulation

The female gametes, i.e. ova, are released from the ovary. The release of the ovum is called ovulation. Ova are not released from the fallopian tube or the uterus; they come out only from the ovary.

5. What is STD?

STD (sexually transmitted diseases) are diseases spread from one person to another through sexual contact, such as gonorrhoea, syphilis and AIDS.

Population Planning

LESSON 24

1. Population and reproduction

Reproduction is necessary for the survival of organisms and the continuation of life, so every generation goes forward. But in humans, when the population grows too much, pressure falls on resources, so population control becomes necessary.

2. Effects of population growth

- Pressure on food increases
- Pressure on water increases
- Shortage of housing increases
- Pressure on education increases
- Pressure on health facilities increases
- Pressure on jobs increases

3. Benefits of control

With population control, resources become enough for everyone and the society becomes successful. When people are fewer and resources are enough, every family easily gets food, water, housing, education, health and jobs.

Understand the series: population growth → pressure on resources → shortage and problems. Its opposite: population control → resources enough → successful society.

Note: for this lesson the direct facts in the source are in one detailed answer, together with the basic points of the importance of reproduction; so the centre of the lesson is this relation of pressure and control.

Inheritance

UNIT 6

Introduction to Inheritance: Chromosomes and Genes

LESSON 25

1. Genetics and inheritance

The branch of biology that studies heredity and variation is called genetics. The way parents transfer their genes to offspring is called inheritance.

2. Chromosome and chromatin

In each cell, each chromosome is a thread-like structure. The centre part of a chromosome, where the two chromatids are joined, is called the centromere. The chemical material of a chromosome (DNA + protein) is chromatin.

3. What is a gene?

The small part of DNA carrying hidden instructions in a language, in which information about making a protein is present, is called a gene. Each gene carries the information for making one protein.

4. Watson and Crick's model of DNA

According to Watson and Crick, DNA is a double helix made of two polynucleotide chains:

- The outer part, i.e. the upright, is made of sugar and phosphate
- The inner rungs are made of nitrogenous bases
- Adenine always pairs with thymine
- Cytosine always pairs with guanine
- The bases are joined by hydrogen bonds

Part of DNA	Make-up
Upright	Sugar and phosphate
Rung	Nitrogenous bases: adenine and thymine, cytosine and guanine

5. Replication and gene expression

Copying of DNA is called replication. A gene expresses itself in two stages: transcription → mRNA is made from DNA; translation → the information in the mRNA is used to make a protein on the ribosome, which expresses the trait.

6. Multiple alleles: human blood groups

Some traits have more than two alternative forms of a gene; these are called multiple alleles. Its best example is the human ABO blood group system, which was discovered by Karl Landsteiner.

- There are four blood groups: A, B, AB and O
- Groups are formed on the basis of the antigens present on the red cells and the antibodies present in the blood serum
- Mixing blood of the wrong group causes agglutination, which is dangerous
- Blood of group O can be given to all groups, so group O is called the universal donor

Law of Segregation

LESSON 26

1. Basic terms

- Allele: different forms of a gene
- Homozygous: both alleles alike, e.g. the same type of trait received from both parents
- Heterozygous: both alleles different
- Dominant allele: the one expressed in the heterozygous condition
- Recessive allele: the one expressed only in the homozygous condition
- Genotype: a trait's genetic makeup
- Homologous: alike chromosomes
- Heterologous: different chromosomes

Term	Relation with dominant and recessive
Dominant trait	Appears in the heterozygous condition, e.g. tall in Tt

Term	Relation with dominant and recessive
Recessive trait	Appears only in the homozygous condition, e.g. short in tt

2. Law of segregation

According to Mendel's law of segregation, the two factors, i.e. alleles, of each trait separate from each other when gametes are formed, and each gamete receives one factor.

3. Monohybrid cross

In a cross of a tall plant (TT) and a short plant (tt), all F1 plants are tall because they are heterozygous (Tt) and the dominant trait appears in them.

Gametes	T	t
T	TT tall	Tt tall
t	Tt tall	tt short

The result of the Tt × Tt cross is: TT, Tt, Tt, tt. So in F2 the ratio of tall to short plants is 3:1 — three parts tall because both TT and Tt have the dominant allele, and one part short.

Series: parents TT × tt → gametes T and t → F1 all Tt tall → cross of F1 among themselves → in F2, 3 tall and 1 short.

Law of Independent Assortment

LESSON 27

1. What is the law of independent assortment?

This is Mendel's second law. According to this law, the genes of different traits are independent of each other in inheritance. When gametes are formed, the genes of one pair assort separately and independently of the genes of another pair.

2. Dihybrid cross

Mendel crossed two traits together: seed colour (yellow dominant, green recessive) and seed shape (round dominant, wrinkled recessive).

- Parents (P): pure yellow round (RRYY) × pure green wrinkled (rryy)
- First generation (F1): all plants were yellow and round, whose genotype is RrYy
- Second generation (F2): when the F1 plants were crossed among themselves, four kinds of plants were produced

3. F2 ratio 9:3:3:1

Kind of plant	Number
Yellow and round	9
Yellow and wrinkled	3
Green and round	3
Green and wrinkled	1

From this ratio it was proved that the gene of colour and the gene of shape assort independently without affecting each other, so new combinations (yellow wrinkled and green round) also appeared in F2.

4. Types of gametes

A plant with genotype RrYy makes four kinds of gametes: RY, Ry, rY and ry. This is independent assortment: in each gamete, one gene from each pair goes independently.

Variation and Evolution

LESSON 28

1. What is variation?

The differences seen in the organisms of a single species are called variation. Variation can be passed to offspring through inheritance. According to Darwin, variations in organisms pass to the next generation through inheritance.

2. Darwin's theory of natural selection

- Organisms produce more offspring than needed
- Among them there is a struggle for existence
- Variation exists in organisms
- The fittest organisms survive
- The surviving organisms pass their traits to offspring generation after generation
- With the passage of time species change; this is evolution

Series: more offspring → struggle for existence → variation → survival of the fittest → transfer of traits → change in species. Organisms with variations suited to the environment survive and produce more offspring, so their traits increase from generation to generation.

3. Comparison of natural and artificial selection

Natural selection	Artificial selection
Takes place through nature	Is done by humans
The fittest organisms survive	Man selects traits of his choice and breeds them

Man and His Environment

UNIT 7

Ecosystem and Ecological Organization

LESSON 29

1. What is the biosphere?

The layer around the earth that sustains life is called the **biosphere**. All organisms live in the biosphere.

2. Levels of ecological organization

- **Population:** a group of organisms of one kind living in one area.
- **Community:** the combined group of different organisms living in an area.
- **Ecosystem:** the area where a community is in relation with the non-living environment; the combined system of a community and the non-living environment is the ecosystem.
- **Biome:** a large biological region recognized by particular plants and a particular climate.
- **Biosphere:** the combined big layer of all ecosystems.

Remember the order from small to big: population → community → ecosystem → biome → biosphere

3. Effect of light

Light is very essential for the ecosystem. Light is essential for photosynthesis; by it plants make food and the whole food chain of the ecosystem runs. Lack of light stops the growth of plants.

4. Biotic components of the ecosystem

The biotic components of an ecosystem are three:

Component	Work	Example
Producers	Make their own food with the sun's energy	Plants
Consumers	Eat plants or other animals	Primary, secondary and tertiary animals
Decomposers	Decay dead organisms and return substances to the soil	Bacteria and fungi

Consumers are of three levels: the first level eats plants, the second level eats the first level, and the third level eats the second level.

Energy Flow in the Ecosystem and Biogeochemical Cycles

LESSON 30

1. What is a food chain?

The food material that passes from the producers among organisms as one eats another is called a **food chain**. A simple example: plant → herbivore → carnivore.

2. Trophic levels

Each step in a food chain is called a trophic level.

- **First trophic level:** the producers (plants), because they make their own food from the sun's energy and feed all other organisms.
- The second level is the primary consumer, the third level the secondary consumer, and so on.

3. Pyramid of numbers

The pyramid of numbers is the chart showing the number of organisms at each trophic level in an ecosystem. Generally the number is greater at the lower levels and smaller at the upper levels.

4. Biogeochemical cycles

The circulation of those elements of the ecosystem that are recycled through organisms is called a **biogeochemical cycle**. The nitrogen cycle and the carbon cycle are its examples.

5. Stages of the nitrogen cycle

Stage	What happens?
Nitrogen fixation	Atmospheric nitrogen changes into compounds that plants can use; this takes place through lightning and nitrogen-fixing bacteria
Nitrification	Nitrogen compounds in organisms change into nitrates; ammonia → nitrates
Uptake by plants	Plants take up nitrates and make protein

The whole cycle runs like this: atmospheric nitrogen → fixation → ammonia → nitrification → nitrates → plants → animals → dead organisms → ammonia, and by denitrification nitrogen goes back into the air.

Interactions and Balance in the Ecosystem

LESSON 31

1. Interactions among organisms

In an ecosystem, organisms are connected with each other in different ways. These interactions keep the balance of the ecosystem.

2. Competition

The cold war between organisms in an ecosystem for food and space is called **competition**. When food or space becomes less, competition takes place among the organisms wanting the same thing.

3. Symbiosis (mutual relationship)

The mutual relationship of two different organisms in which both benefit is called **symbiosis**. In its special form, neither can live without the other.

- Example: in **lichens**, algae and fungi live together and both benefit each other.
- Such a mutually beneficial relationship is also linked with mutualism.

4. Other interactions to remember

Interaction	Meaning
Competition	Rivalry / cold war for food and space
Symbiosis	Both benefit, living without each other is difficult
Predation	One organism eats another
Commensalism	One benefits, the other neither benefits nor is harmed

5. Why is balance necessary?

If one kind of organism becomes too many or too few, the food chain breaks and the balance of the ecosystem is disturbed. Competition and mutual interactions help to keep numbers in balance.

Human Effects on the Environment

LESSON 32

1. Human activities and the environment

Human activities have a big effect on the environment. The increase of gases, the spoiling of water and the ending of organisms are its results.

2. Greenhouse effect

The **greenhouse effect** is the trapping of the sun's heat on the earth by gases like carbon dioxide, which raises the earth's temperature. With rising temperature the climate changes.

3. Algal bloom

Algal bloom is the rapid growth of algae in water.

- Algae grow rapidly and use up the water's oxygen.
- Without oxygen, the water organisms die.
- In this way the ecosystem is destroyed.

Remember the flow: rapid growth of algae → oxygen used up → water organisms die → ecosystem destroyed

4. Endangered species

Some organisms are in danger of ending. The endangered species found in Pakistan include:

- Markhor
- Snow leopard
- Indus dolphin
- Green turtle
- Houbara bustard (talor)

Protecting these species is everyone's responsibility, because if one species ends, the food chain and balance are disturbed.

Pollution and Conservation of the Natural Environment

LESSON 33

1. What is pollution?

Pollution is the addition of harmful substances to the environment. Pollution spoils air, water and soil — all three.

2. Air pollution

The increase in the amount of harmful gases in the air besides oxygen is called **air pollution**.

Type of pollution	Harmful effect
Air pollution	Breathing diseases, asthma, bronchitis, lung cancer, irritation of the eyes
Acid rain (a result of air pollution)	Damage to crops
Damage to the ozone layer	More harmful rays of the sun reach us
Water pollution	Water organisms die, algal bloom takes place, drinking water is spoiled

3. Steps to stop water pollution

- Factory waste water should be cleaned before releasing into rivers.
- Garbage should not be thrown into water.
- Less fertiliser and pesticide should be used in fields.
- Water should be saved, not wasted.

4. Conservation of the natural environment

To save the environment, pollution must be reduced, endangered species must be protected, and water and air must be kept clean. Only in a clean environment do the balance of the ecosystem and human health remain.

Biotechnology

UNIT 8

Introduction to Biotechnology and the Fermentation Industry

LESSON 34

1. What is biotechnology?

Biotechnology is the preparation of useful products and services for humans using biological systems, organisms or their parts.

2. Traditional biotechnology

Traditional biotechnology is the preparation of food and other products using organisms or their parts, such as making bread and curd by fermentation.

- Biotechnology is an **ancient** technology: for thousands of years humans have made bread, curd, cheese and vinegar by fermentation.
- Selecting better crops and animals by breeding is also ancient biotechnology.
- The early biotechnologists were **farmers**, who made things like curd, bread and wine by fermentation.

3. Services of Louis Pasteur

- Proved that fermentation is caused by living organisms (yeast).
- Gave the method of pasteurization.
- Laid the basis for vaccine preparation.

4. Industrial biotechnology and the fermenter

Large-scale preparation of alcohol and antibiotics through organisms is a special part of **industrial biotechnology**. The vessel used for large-scale production of bacteria is called a fermenter.

5. How does fermentation work?

Organisms produce energy (ATP) by breaking down food. Most organisms use oxygen to produce ATP. In fermentation glucose breaks down without oxygen:

Glucose → pyruvic acid → alcohol or lactic acid + a little energy

- Organisms that carry out fermentation: yeast, some bacteria (such as Lactobacillus) and the muscles of animals in the absence of oxygen.
- Factors affecting it: temperature, pH, presence of oxygen, amount of food (sugar) and type of organism.

6. Types of fermentation

Type	Product
Alcoholic fermentation	Alcohol + carbon dioxide; the released carbon dioxide makes the dough rise
Lactic acid fermentation	Lactic acid; lactic acid is formed from pyruvic acid
Acidic fermentation	Acids like vinegar; lactic acid is formed from pyruvic acid

7. Products, advantages and disadvantages

Fermentation produces curd, cheese, bread, alcohol, vinegar, antibiotics and organic acids. Pakistani foods made through fermentation include curd, lassi, pickles, vinegar, leavened bread (naan, double roti) and cheese.

- **Lactobacillus:** a bacterium which converts the sugar of milk (lactose) into lactic acid, so the milk sets into curd.
- **Advantages:** food becomes tasty and easy to digest, keeps longer, and gains nutritional value.
- **Disadvantages:** bad fermentation can spoil food and produce harmful substances.

Genetic Engineering

LESSON 35

1. What is genetic engineering?

Artificially changing/modifying/rearranging DNA is called **genetic engineering**. By it, the gene of one organism is put into another organism and a new trait is produced.

2. Tools of recombinant DNA

Tool / enzyme	Work
Restriction enzyme	Cuts DNA; bacteria naturally use it for their defence, they cut the DNA of viruses
Ligase enzyme	Rejoins the cut DNA
Plasmid	Extra DNA used as a vector (carrier) for a gene

3. Steps of genetic engineering

Taking the desired gene → cutting DNA with a restriction enzyme → putting the gene into a plasmid → joining with ligase → putting the recombinant DNA into bacteria → production of the new product

4. Human Genome Project (HGP)

The study of the complete map of the human genome is the **Human Genome Project (HGP)**. By it, the sequence and place of all human genes were found.

5. Recombinant products in the market

- Insulin
- Human growth hormone
- Hepatitis B vaccine
- Interferon

These products are obtained from recombinant DNA technology and are available in the market.

Single Cell Protein and Its Uses

LESSON 36

1. What is single cell protein (SCP)?

Single cell protein is the protein obtained from the cells of microscopic organisms. This term was given in 1966 by Professor C. C. Wilson of the Massachusetts Institute of Technology.

2. What does SCP contain?

- 60 to 80 percent of the dry weight of the cells is protein
- It also contains nucleic acids, fats (lipids), minerals and vitamins
- Some kinds have a higher amount of amino acids

3. Which organisms and which food (substrate)?

Thing	Examples
Organisms	Bacteria, yeast, fungi, algae
Carbon substrates	n-alkanes, alcohol products, sulphite, molasses, whey (milk water)
Place of production	Grown inside a fermenter on a large scale

4. Uses and advantages

- Single cell protein is included in the food of very many animals; it is especially used as animal food and as a nutritional supplement
- Some waste substances can be made useful by fermentation, e.g. products can be made using grass, wood or straw
- Leftover waste, leftover alcohol and human and animal waste can also be used as substrate, which also reduces the waste problem

Pharmacology

UNIT 9

Introduction to Pharmacology and Medicines for Diseases

LESSON 37

1. Pharmacology and sources of medicines

Medicines are obtained from different natural sources. Medicines from natural sources are obtained by extracting and purifying substances from plants, animals and microorganisms.

Source	Parts / examples
Plants	Leaves, bark, roots; the medicine for Rheumatoid Arthritis (aspirin-like drug) is obtained from plants
Animals	Glands, liver; insulin (from the pancreas of cow/pig), heparin (from the liver), cod liver oil (from fish liver)
Microorganisms	Medicines are extracted and purified from fungi and bacteria
Minerals	Minerals are also a source of medicines

2. Founder of antiseptic

Joseph Lister is considered the founder of antiseptic. With antiseptics, germs in wounds are killed and infection is prevented.

3. Names to remember

- Lister — founder of antiseptic
- Alexander Fleming, Edward Jenner and Louis Pasteur are other famous names, but the founder of antiseptic is Lister.

Addictive Drugs

LESSON 38

1. Addictive drugs

Some drugs have such an effect on the brain and body that a person becomes addicted to them. Such a habit is called drug addiction and it is very dangerous.

Type of drug	Effect
Sedatives	Slow down brain activity and bring sleep
Hallucinogens	Effects like seeing imaginary things and sudden happiness
Narcotics	One type of addictive drugs
Marijuana	Counted among addictive substances

2. Why is the use of sedatives increasing?

- Stress (tension) has increased.
- Anxiety has increased.
- Complaints of sleeplessness have increased.
- People rely on them for rest and sleep and then become addicted to them.

3. Why is drug addiction dangerous?

- Addiction damages the brain and the body.
- Health is destroyed.
- Problems are created in the family and society.
- A person becomes a slave of the drug.

Therefore addiction should be avoided and such medicines should not be used without a doctor's advice.

Antibiotics and Vaccines

LESSON 39

1. Antibiotics

Antibiotics are medicines that work against bacteria.

- **Bacteriostatic:** things that stop the growth of bacteria but do not kill them.
- **Bactericidal:** the word connected with the property of killing bacteria.

2. Why don't antibiotics work against viruses?

Antibiotics act on the walls and processes of bacteria; viruses lack those structures and live inside cells, so antibiotics do not affect them.

3. Misuse of antibiotics and its harm

- Destruction of useful bacteria
- Allergy
- Diarrhoea
- **Antibiotic resistance** — the biggest result of misuse

4. How do bacteria develop resistance?

A resistant trait is produced by mutation → bacteria make enzymes that break down the antibiotic, pump it out, or change their target → resistant bacteria survive and multiply, and resistance spreads to other bacteria.

5. Vaccine and vaccination

A vaccine is given **before disease** so that immunity develops in the body.

- In vaccination the vaccine is given into the body by injection or drops.
- It contains weakened or dead germs which do not cause disease.
- The body recognises them and makes antibodies and memory cells.
- When the real germs arrive, the body destroys them at once.

Flow: vaccine (weakened/dead germs) → recognition by the body → antibodies + memory cells → on arrival of real germs, immediate destruction → protection from disease

Thing	When / work
Vaccine	Before disease, for immunity
Antibiotic	In bacterial disease, ineffective against viruses
Bacteriostatic	Stops the growth of bacteria, does not kill